V93000 Specifications

PMUX01 Power MUX Card

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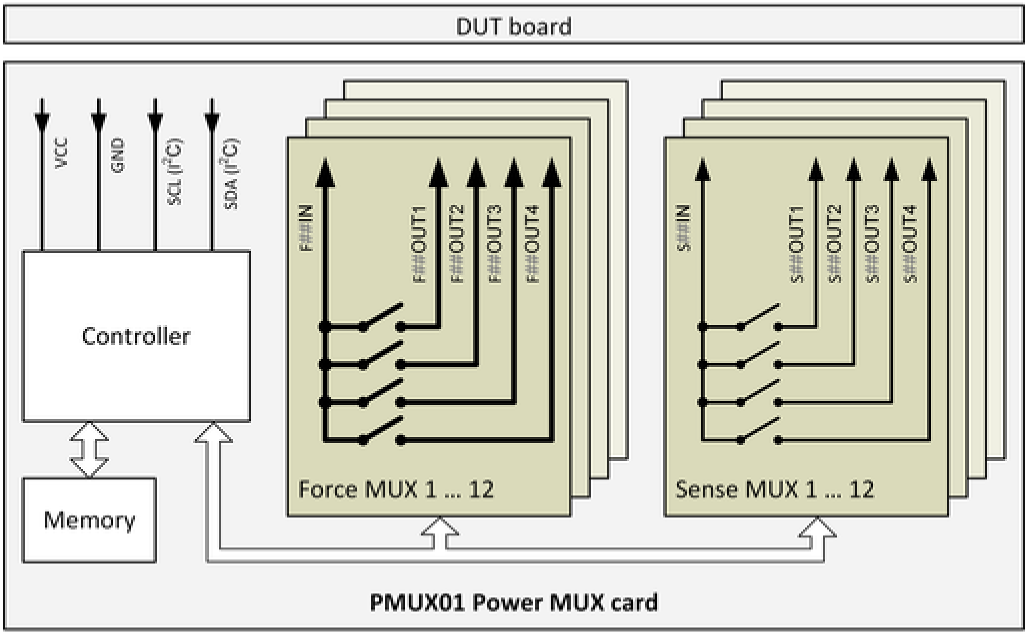
PMUX01 Power MUX card

The specification covers the PMUX01 Power MUX (Part Number E8021PM)

Configuration and topology

Total force switches	48
Total sense switches	48
Force switch topology	12 x 1:4 (1 IN, 4 OUT)
Sense switch topology	12 x 1:4 (1 IN, 4 OUT)
Maximum card count per system	16 (max. 12 for a Direct-Probe™ configuration)



PMUX01 Power MUX block diagram

Power supply

Card supply	External power supply option via DUT board
Card supply voltage	+5 V ± 0.25 V
Maximum current consumption (idle)	100 mA
Maximum current consumption per force switch ON	8 mA
Maximum current consumption per sense switch ON	5 mA
Maximum total current consumption (ALL switches ON)	700 mA

Force switch specifications

Typical path resistance (IN to OUT)	100 mΩ
Maximum path resistance (IN to OUT)	250 mΩ
Maximum voltage across open switch (IN to OUT)	±120 V
Maximum voltage between adjacent pogo pins	±160 V

Maximum voltage to DUT board ground	±120 V
Maximum continuous current (IN, OUT)	1 A
Maximum pulse current (IN, OUT)	5 A, maximum 10 ms, 10 % duty cycle
Typical OUT to IN leakage current @ 40V[1]	10 nA
Typical OUT to IN leakage current @ 80V[1]	20 nA
Maximum OUT to IN leakage current[1]	11 nA + 1 nA / V[2]
Maximum IN to OUTx leakage current[3]	46 nA + 4 nA / V[4]
Switch on time	0.2 ms typical, 0.5 ms maximum
Switch off time	0.2 ms typical, 0.5 ms maximum
Coupling capacity (IN-OUT)	1000 pF

Sense switch specifications

Typical path resistance (IN to OUT)	20 Ω
Maximum path resistance (IN to OUT)	40 Ω
Maximum voltage across open switch (IN to OUT)	±120 V
Maximum voltage between adjacent channels	±160 V
Maximum voltage to DUT board ground	±120 V
Maximum continuous current	0.1 A
Typical OUT to IN leakage current @ 40V[1]	4 nA
Typical OUT to IN leakage current @ 80V[1]	8 nA
Maximum OUT to IN leakage current[1]	6 nA + 0.2 nA / V[5]
Maximum IN to OUTx leakage current[3]	18 nA + 0.8 nA / V[6]
Switch on time	0.2 ms typical, 1 ms maximum
Switch off time	0.2 ms typical, 1 ms maximum
Coupling capacity (IN-OUT)	100 pF

Control specifications

- On-board controller handles the data transfer via the I^{2}C interface, interprets the received commands and applies the settings.
- The I^{2}C interface complies with the I2C-bus Specification, Rev. 5. The Fast-mode Plus (Fm+) with a bit rate up to 1 Mbit/s is supported.
- Four addressing bits on pogo pins ID0 to ID3 select one of the maximum 16 cards.
- On-board volatile memory stores complete setups to minimize the programming time. The memory stores up to 8192 complete setups.

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[1] Leakage current at one OUT pin with the IN pin connected to ground and other OUT pins open.
[2] Tested with test limits of 25 nA + 1 nA / V. The test limits include a measurement deviation of 14 nA.
[3] Leakage current at the IN pin with all OUT pins connected to ground.
[4] Tested with test limits of 60 nA + 4 nA / V. The test limits include a measurement deviation of 14 nA.
[5] Tested with test limits of 20 nA + 0.2 nA / V. The test limits include a measurement deviation of 14 nA.
[6] Tested with test limits of 32 nA + 0.8 nA / V. The test limits include a measurement deviation of 14 nA.

Other configuration data

- The DUT board stiffener MUXSTIF01 is required for applications that connect to the PMUX01 Power MUX card. For more information, refer to topic 146875 - "MUX card stiffeners overview".
- The interposer MUXIP01 is required for applications that connect to the card. The interposer covers two PMUX01 Power MUX cards and is mounted on the bottom side of the DUT board. For more information, refer to topic 146845 - "MUX Cards Interposer (MUXIP01)".
- Diagnostics requires the MUXDIAG (E7010MUX) Diagnostics Interconnect Board.
- The Enhanced DUT Scale Interface is required.